

CVSS-X: A Multilingual Speech-to-Speech Translation Corpus for 28 Languages

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Abstract

We introduce CVSS-X, a large-scale synthetic speech-to-speech translation corpus that extends CVSS by reversing the translation direction. While CVSS translates from 21 languages into English, CVSS-X enables translation from English into 28 target languages spanning 12 language families. The corpus comprises approximately 240,000 parallel speech pairs per language, totaling over 16,000 hours, eight times larger than CVSS. We provide two variants: CVSS-X-C with two canonical voices per language, and CVSS-X-T with cross-lingual voice cloning, both fully generated. Evaluation shows comparable translation quality to CVSS with consistent performance across typologically diverse languages. Combined with CVSS, this enables research on bidirectional and multilingual speech-to-speech translation. The code is available at <https://github.com/ErmiAI/XVSS-X> and the dataset under CC-BY-NC 4.0 license at <https://huggingface.co/datasets/lgris/XVSS-X>.

1 Introduction

Speech-to-speech translation (S2ST) is a key technology for breaking down communication barriers between speakers of different languages. Traditionally, S2ST systems rely on a cascade of automatic speech recognition (ASR), text-based machine translation (MT), and text-to-speech (TTS) synthesis (Waibel et al., 1991). Recently, direct S2ST approaches that bypass intermediate text representations have gained significant attention, including end-to-end models such as Translatotron (Jia et al., 2019, 2022a) and methods based on discrete speech units (Lee et al., 2022; Popuri et al., 2022).

The release of the CVSS corpus (Jia et al., 2022b) marked a turning point, providing the first large-scale, publicly available parallel speech corpus and enabling substantial progress in direct S2ST research. However, CVSS is restricted to a many-to-one setting, translating from 21 source languages exclusively into English. This limitation constrains the development of more inclusive multilingual S2ST technologies, as researchers working on translation *from* English or between non-English language pairs still face a scarcity of suitable training data.

In this paper, we introduce **CVSS-X**, a massively multilingual speech-to-speech translation corpus that complements and extends the original CVSS by reversing the translation direction. While CVSS translates *into* English from 21 source languages, CVSS-X translates *from* English into 28 target languages spanning 12 language families, including Germanic, Romance, Slavic, CJK, and others. CVSS-X is derived from English Common Voice recordings (Ardila et al., 2020), with translations generated using NLLB-200 (Costa-jussà et al., 2022) and synthesized via OmniVoice (Zhu et al., 2026), a state-of-the-art (SOTA) multilingual TTS system with voice cloning capabilities.

Following the design of CVSS, we provide two variants: **CVSS-X-C** (Canonical), where target speech is synthesized using two fixed reference voices per language, and **CVSS-X-T** (Transferred), where the voice characteristics of the original English speaker are preserved through cross-lingual voice cloning. The combination of CVSS and CVSS-X enables, for the first time, large-scale research on bidirectional and multilingual S2ST, supporting translation not only to and from English, but also between arbitrary language pairs using English as a pivot.

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Table 1: Comparison of S2ST corpora. Dirs: number of supported translation directions. Type: *Real* (human recordings), *Mined* (automatically aligned), *Synth* (TTS-generated).

Dataset	Dirs	Langs	Hours	Type	License
Fisher ES-EN	2	2	127	Real	LDC
VoxPopuli	210	15	17.3K	Real	CC0
SpeechMatrix	136	17	418K	Mined	CC-BY-NC
SeamlessAlign	–	37	29K	Mined	CC-BY-NC [†]
CVSS-C/T	21	21	1.9K	Synth	CC-BY
CVSS-X-C/T	28	28	16K	Synth	CC-BY-NC

[†] Metadata only; requires reconstruction from Common Crawl.

2 Related Work

Table 1 summarizes publicly available corpora for S2ST research. True parallel speech corpora remain rare due to the high cost of collecting aligned utterances across languages. CVSS (Jia et al., 2022b) was the first large-scale public S2ST corpus, but is limited to many-to-one translation (X→EN) with 21 source languages. SpeechMatrix (Duquenne et al., 2023) provides massive scale (418K hours) through automatic speech mining from European Parliament recordings, though it covers only 17 European languages and alignment quality is inherently approximate. The SeamlessM4T project (Seamless Communication et al., 2023) introduced SeamlessAlign, expanding coverage to 37 languages, but only metadata is publicly released, requiring users to reconstruct the dataset from Common Crawl archives.

CVSS-X addresses these limitations by providing a fully downloadable synthetic corpus that enables one-to-many translation (EN→X) across 28 languages from 12 typological families (grouped into 7 macro-categories). Unlike mined corpora, CVSS-X offers perfect sentence-level alignment since source and target speech are generated from verified parallel text.

3 CVSS-X Dataset

3.1 Data Sources

Source Data. We construct CVSS-X using English source utterances from Common Voice version 17 (Ardila et al., 2020). To establish direct symmetry with CVSS (Jia et al., 2022b), which synthesized English target speech corresponding to CoVoST 2 English transcripts, we align Common Voice version 17 recordings with the original Common Voice version 4 utterances used by CVSS/CoVoST 2 via normalized text matching. This recovered

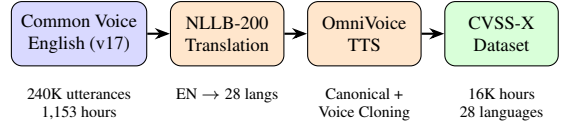


Figure 1: CVSS-X dataset creation pipeline.

240,192 of the original 264,037 English utterances (91.0%) with verified human source audio. Since the matched dev split contained only 871 samples, we supplemented it by randomly sampling from the training split to reach exactly 10,000 samples. The final splits are: **train** (222,349), **dev** (10,000), and **test** (7,843) samples per target language.

Text Translation.

Source transcripts are translated using the distilled NLLB-200 model¹ (Costa-jussà et al., 2022), selected after benchmarking seven candidate translation models on a closed evaluation set. Translation quality was assessed through an LLM-as-a-judge protocol over 100 EN→PT samples, with each output rated on a 1–10 scale for accuracy, fluency, and terminology preservation.

NLLB-200 was ultimately selected due to its strong performance in this evaluation, together with its broad coverage of languages and language families, making it particularly suitable for the multilingual setting considered in this work.

Speech Synthesis. Target speech is synthesized using OmniVoice (Zhu et al., 2026), a SOTA multilingual TTS system with zero-shot voice cloning capabilities. OmniVoice was selected for its ability to perform cross-lingual voice cloning, synthesizing speech in a target language from a voice reference spoken in a different source language. This capability is particularly important for CVSS-X-T, where the available voice reference is English speech, as it helps preserve speaker identity while minimizing the transfer of English-specific accent characteristics to the synthesized target-language speech.

3.2 Generation Pipeline

Figure 1 illustrates the CVSS-X generation pipeline. We generate two corpus variants:

CVSS-X-C (Canonical): Two fixed reference voices per language (one male, one female), created using ElevenLabs’² voice design feature, with synthetic speakers with neutral accents op-

¹<https://huggingface.co/facebook/nllb-200-distilled-600M>

²<https://elevenlabs.io>

Table 2: CVSS-X target languages by family.

Family	N	Languages
Romance	6	PT, ES, FR, IT, RO, CA
Germanic	5	DE, NL, SV, DA, NO
Slavic	4	RU, PL, CS, UK
CJK	3	ZH, JA, KO
Uralic	2	FI, HU
Indo-Iranian	2	HI, FA
Other	6	EL, HE, TR, TH, ID, VI

Table 3: CVSS-X corpus statistics.

Metric	Value
Source language	English
Target languages	28
Language families	7
Samples per language	240,192
Total translated pairs	6,725,376
CVSS-X-C	
Duration	~6,730 hours
Avg. utterance	3.6 seconds
CVSS-X-T	
Duration	~9,340 hours
Avg. utterance	5.0 seconds
Total duration	~16,070 hours

timized for clarity. Voice selection is based on source speaker gender metadata: samples with gender=male, empty, or other use the male voice; gender=female samples use the female voice. This results in 81.4% male and 18.6% female distribution, doubling voice diversity compared to CVSS-C’s single canonical voice.

CVSS-X-T (Timbre-transferred): Zero-shot voice cloning conditioned on the source English audio, preserving speaker characteristics across languages. Approximately 3.4% of samples are skipped due to insufficient signal for voice cloning.

3.3 Target Languages

CVSS-X covers 28 languages spanning 12 distinct phylogenetic and typological language families (Romance, Germanic, Slavic, Sinitic, Japonic, Koreanic, Uralic, Indo-Iranian, Hellenic, Semitic, Turkic, and Austroasiatic/Austronesian/Kra-Dai), grouped into 7 macro-categories in Table 2 to facilitate comparative analysis.

3.4 Corpus Statistics

Table 3 summarizes corpus statistics. Both CVSS-X-C and CVSS-X-T are fully generated, totaling over 16,000 hours of parallel speech.

Table 4: Evaluation by language family (CVSS-X).

Family	WER/CER		ASR-BLEU		UTMOS	
	C	T	C	T	C	T
Romance (6)	5.8	8.2	90.3	88.0	3.54	3.27
Germanic (5)	9.5	12.2	85.3	81.1	3.62	3.27
Slavic (4)	7.0	7.7	86.9	85.9	3.48	3.17
CJK (3)*	5.0	10.2	75.2	67.9	3.56	3.20
Uralic (2)	10.3	14.3	84.0	78.3	3.54	3.21
Indo-Iranian (2)	22.5	23.0	63.7	60.0	3.61	3.20
Other (6)	24.8	24.8	65.0	64.9	3.53	3.14
Average	12.1	14.1	82.4	79.4	3.55	3.21

*CER reported; BLEU w/ specific tokenizers.

4 Dataset Quality Evaluation

We evaluate CVSS-X quality and compare with the original CVSS corpus. Note that direct comparison has limitations since CVSS evaluates $X \rightarrow EN$ while CVSS-X evaluates $EN \rightarrow X$.

4.1 Evaluation Setup

We evaluate a stratified random sample of 200 utterances per language from the dev set (5,600 samples per variant). This sample size was determined via power analysis: given observed standard deviations of $\sigma \approx 0.5$ for UTMOS, 200 samples yields 95% confidence intervals of ± 0.07 , sufficient to detect meaningful differences. To evaluate the acoustic intelligibility and text preservation of the synthesized target speech, we perform Whisper large-v3 ASR (Radford et al., 2023) and compute Word/Character Error Rate (WER/CER) and ASR-BLEU against the translated text prompt (measuring TTS-to-ASR round-trip fidelity). For unsegmented languages (ZH, JA, TH), standard whitespace-based SacreBLEU produces zero; we therefore employ language-specific tokenizers (*jieba*, *fugashi*, and *pythainlp*) and report Character Error Rate (CER) and chrF2++ (Popović, 2015, 2016, 2017). Naturalness is measured with UTMOS (Saeki et al., 2022), and speaker similarity with ECAPA-TDNN (Desplanques et al., 2020). To enable fair comparison with CVSS, we re-evaluate the original corpus using the same models and sample size.

4.2 Results

Table 4 shows results by language family, and Table 5 provides an overall comparison.

TTS quality gap. CVSS-X-C achieves lower UTMOS (3.55) compared to CVSS-C (4.43). This gap stems from task complexity: CVSS synthesizes only English using PnG NAT trained on high-quality LibriTTS data, while CVSS-X must synthesize 28 typologically diverse languages with a

Table 5: Overall comparison with CVSS (re-evaluated with same pipeline).

Metric	X-C	X-T	CVSS-C	CVSS-T
UTMOS (1–5)	3.55	3.21	4.43	3.61
ASR-BLEU	82.4	79.4	94.2	93.8
WER/CER (%)	12.1	14.1	3.5	4.0
Spk. Similarity	–	0.607	–	–

single multilingual model. The consistent UTMOS across language families (3.48–3.62) suggests OmniVoice provides uniform quality rather than excelling in some languages at the expense of others.

Voice cloning quality. CVSS-X-T achieves UTMOS of 3.21, a modest reduction from CVSS-X-C (3.55). This gap is smaller than expected given that OmniVoice preserves acoustic characteristics from the source audio, including potential artifacts from crowdsourced recordings. Speaker similarity of 0.607 indicates good voice preservation, with Germanic languages achieving higher similarity (0.648) than Slavic (0.567), possibly due to phonetic proximity to English.

TTS intelligibility and fidelity. Romance and Slavic families achieve the highest BLEU scores (88–90) and lowest error rates (WER < 8%), demonstrating high synthesis fidelity across OmniVoice. Indo-Iranian and Other families show lower scores (60–65) driven by non-Latin scripts such as Hebrew (ASR-BLEU=46.1) and complex tonal dynamics in Thai. When using language-specific word tokenizers, unsegmented languages achieve strong fidelity: Thai reaches word-tokenized BLEU of 49.1 (C) and 47.8 (T), while Chinese (ZH) and Japanese (JA) achieve tokenized BLEU of 87.6/70.4 and 88.9/85.7, respectively. The higher WER in the “Other” family (24.8%) is primarily driven by Thai (77.0% WER without tone diacritics in Whisper) and Hebrew (37.4% WER).

5 Conclusion

We introduced CVSS-X, a large-scale synthetic S2ST corpus that reverses the translation direction of CVSS, enabling EN→X translation into 28 target languages across 12 language families. Both variants are fully generated: CVSS-X-C (6,730 hours) and CVSS-X-T (9,340 hours), totaling over 16,000 hours of parallel speech—eight times larger than CVSS. Combined with CVSS, this enables bidirectional translation and, through English as pivot, arbitrary X→Y language pairs.

Future versions will address current limitations: regenerating translations with TranslateGemma-12B would improve fidelity and enable Apache 2.0 licensing; incorporating Common Voice v26 (significantly more speakers) and Common Voice Spontaneous Speech 4.0 (natural conversational recordings) would increase diversity and scale while providing more realistic voice cloning scenarios. We also plan to train and evaluate baseline S2ST models to validate the dataset.

Limitations

The current release has several limitations: (1) translation quality depends on NLLB-200, which, despite verified fidelity, may introduce errors for low-resource languages; (2) while ASR-BLEU measures TTS acoustic intelligibility against prompt text, cross-validation against human-authored references (such as CoVoST 2 English-to-15 subsets) remains an ongoing benchmark; (3) the CC-BY-NC license (inherited from NLLB-200) restricts commercial applications; (4) comparison between CVSS (X→EN) and CVSS-X (EN→X) is inherently asymmetric; and (5) automated MOS predictors (UTMOS) may exhibit variance across non-Western phonologies, necessitating human perceptual validation in future iterations.

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